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Large Language Models #1



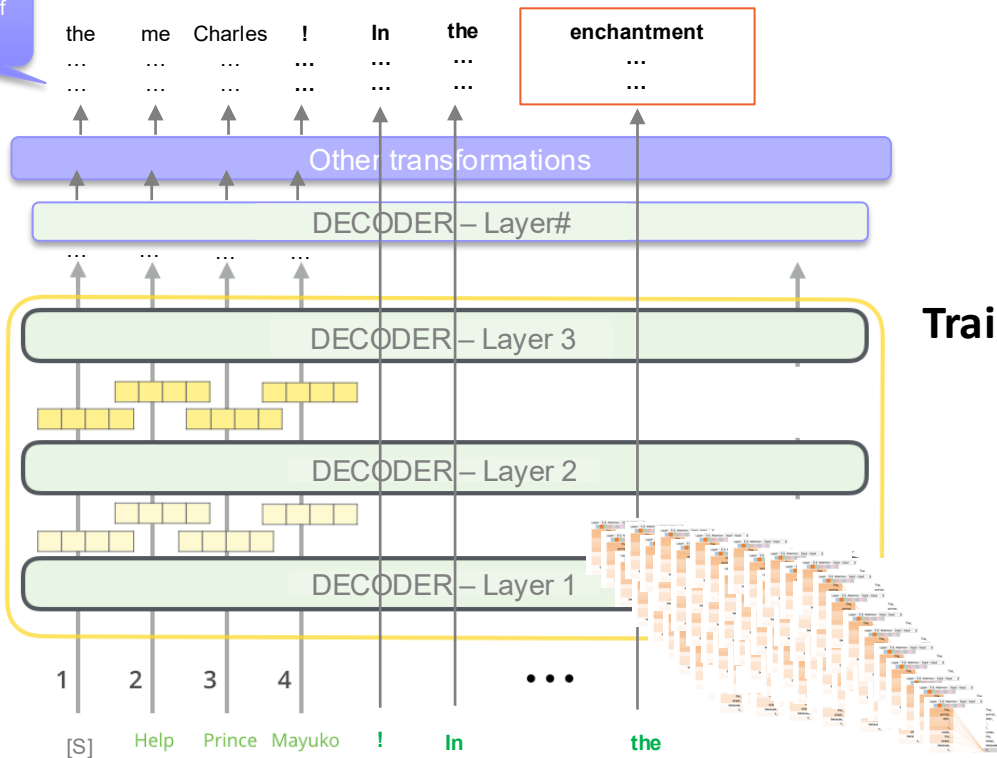
LARGE LANGUAGE MODELS (LLM)

Parameters, scaling laws, in-context learning,
additional training methods

Parameters in GPT3

Probability of each token

X 96



GPT3 185B

- 96 layers; each one...
 - 96 attention heads
 - + Feed forward network matrices
- Token-level representation dimension:
 - 12288

Training data for similar model (LLAMA)

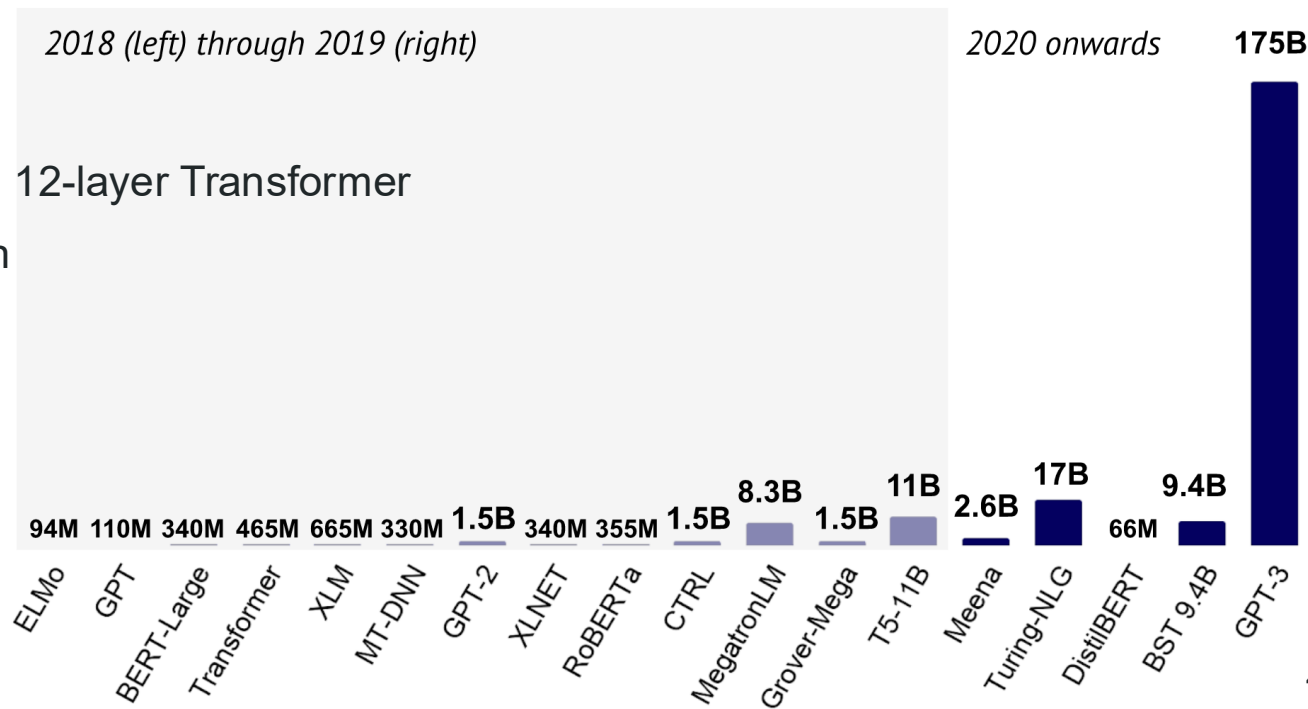
Dataset	Sampling prop.	Epochs	Disk size
CommonCrawl	67.0%	1.10	3.3 TB
C4	15.0%	1.06	783 GB
Github	4.5%	0.64	328 GB
Wikipedia	4.5%	2.45	83 GB
Books	4.5%	2.23	85 GB
ArXiv	2.5%	1.06	92 GB
StackExchange	2.0%	1.03	78 GB

Model Name	n_{params}	n_{layers}	d_{model}	n_{heads}	d_{head}	Batch Size	Learning Rate
GPT-3 Small	125M	12	768	12	64	0.5M	6.0×10^{-4}
GPT-3 Medium	350M	24	1024	16	64	0.5M	3.0×10^{-4}
GPT-3 Large	760M	24	1536	16	96	0.5M	2.5×10^{-4}
GPT-3 XL	1.3B	24	2048	24	128	1M	2.0×10^{-4}
GPT-3 2.7B	2.7B	32	2560	32	80	1M	1.6×10^{-4}
GPT-3 6.7B	6.7B	32	4096	32	128	2M	1.2×10^{-4}
GPT-3 13B	13.0B	40	5140	40	128	2M	1.0×10^{-4}
GPT-3 175B or “GPT-3”	175.0B	96	12288	96	128	3.2M	0.6×10^{-4}

Language Model “Scaling Wars”

BERT-base: 110M params, 12-layer Transformer

GPT-4: GPT-3*6 ~ 1 Trillion



Zero-shot

The model predicts the answer given only a natural language description of the task. No gradient updates are performed.

```
1 Translate English to French: ← task description
2 cheese => ..... ← prompt
```

One-shot

In addition to the task description, the model sees a single example of the task. No gradient updates are performed.

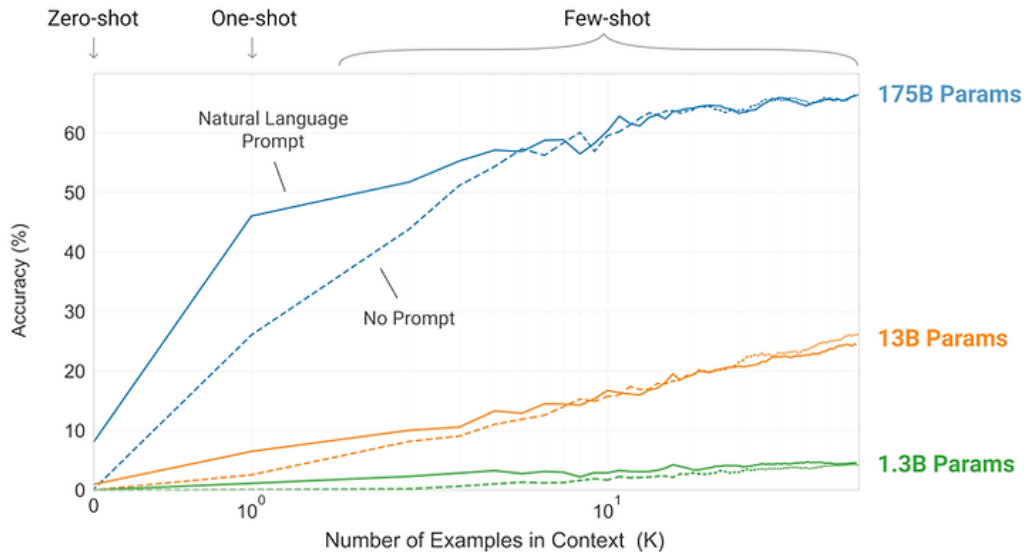
```
1 Translate English to French: ← task description
2 sea otter => loutre de mer ← example
3 cheese => ..... ← prompt
```

Few-shot

In addition to the task description, the model sees a few examples of the task. No gradient updates are performed.

```
1 Translate English to French: ← task description
2 sea otter => loutre de mer ← examples
3 peppermint => menthe poivrée ←
4 plush girafe => girafe peluche ←
5 cheese => ..... ← prompt
```

Language Model “Scaling Wars” and In-context Learning



OpenAI: *Language Models are Few-Shot Learners*

<https://arxiv.org/pdf/2005.14165.pdf>

See also: <https://www.theaidream.com/post/openai-gpt-3-understanding-the-architecture>

And: https://people.cs.umass.edu/~miyyer/cs685/slides/prompt_learning.pdf

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In-context Learning (LLMs)

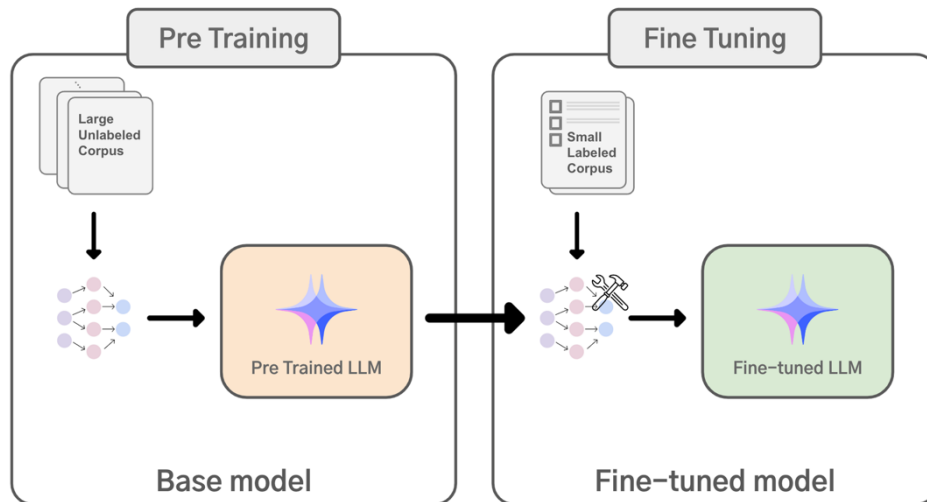
- A new learning paradigm at inference time:
 - A few “training data” in the prompt, also called **demonstrations**
 - Improve downstream performance
 - Help learn hidden patterns from examples: why?
- Adaptation via prompting
 - Increase generalization and the capability to handle different tasks

... but ... the behavior of “raw” (generative) LLMs trained in self-supervised fashion is still limited

SELF-SUPERVISED PRE-TRAINING

VS.

FINE-TUNING



Source: <https://www.upstage.ai/feed/tech/understanding-finetuning>

InstructGPT: Alignment of LLM and User Preferences

GPT3
→ GPT3.5

GPT3.5 =
GPT3 +
InstructGPT

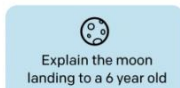
~ “ChatGPT”

SFT:
Supervised Fine-tuning

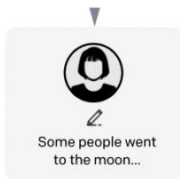
Step 1

**Collect demonstration data,
and train a supervised policy.**

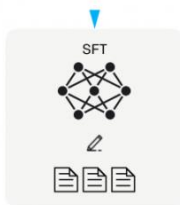
A prompt is
sampled from our
prompt dataset.



A labeler
demonstrates the
desired output
behavior.



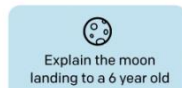
This data is used
to fine-tune GPT-3
with supervised
learning.



Step 2

**Collect comparison data,
and train a reward model.**

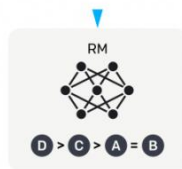
A prompt and
several model
outputs are
sampled.



A labeler
ranks the outputs
from best to worst.



This data is used
to train our
reward model.



Step 3

**Optimize a policy against
the reward model using
reinforcement learning.**

iq.opengenus.org

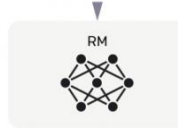
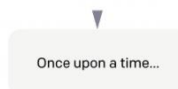
A new prompt
is sampled from
the dataset.



The policy
generates
an output.



The reward model
calculates a
reward for
the output.



The reward is
used to update
the policy
using PPO.



RLHF:
Reinforcement
Learning with
Human
Feedback

Example of Data for Supervised Fine-Tuning #1

instruction string · lengths	input string · lengths	output string · lengths	text string · lengths
 9-58 51.7%	 0-247 98.8%	 0-419 74.5%	 154-589 67.2%
Render a 3D model of a house		<nooutput> This type of instruction cannot be...	Below is an instruction that describes a task. Write a...
Evaluate this sentence for spelling and grammar mistakes	He finnished his meal and left the resturant	He finished his meal and left the restaurant.	Below is an instruction that describes a task, paired with an input that provides further context. Write a response that appropriately completes the request. ### Instruction: Evaluate this sentence for spelling and grammar mistakes ### Input: He finnished his meal and left the resturant ### Response: He finished his meal and left the restaurant.

Example from Stanford's Alpaca dataset:

<https://huggingface.co/datasets/tatsu-lab/alpaca/viewer/default/train?views%5B%5D=train&row=9>

Example of Data for Supervised Fine-Tuning #2

instruction string · lengths	input string · lengths	output string · lengths	text string · lengths
 9-58 51.7%	 0-247 98.8%	 419-838 22.1%	 589-1.02k 28.2%
Discuss the causes of the Great Depression		The Great Depression was a severe global economic downturn that lasted from 1929 to 1939. It was caused by several factors, including the stock market crash of 1929, structural weaknesses in the economy, low consumer spending, high unemployment, and a lack of confidence in the banking system. Other contributing factors included overproduction of commodities, reliance on borrowing to finance investments, agricultural overproduction, and a decrease in international trade.	Below is an instruction that describes a task. Write a response that appropriately completes the request. ### Instruction: Discuss the causes of the Great Depression ### Response: The Great Depression was a severe global economic downturn that lasted from 1929 to 1939. It was caused by several factors, including the stock market crash of 1929, structural weaknesses in the economy, low consumer spending, high unemployment, and a lack of confidence in the banking system. Other contributing factors included overproduction of commodities, reliance on borrowing to finance investments, agricultural overproduction, and a decrease in international trade.

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Collecting Preference Data for RLHF

Pick the best answer to improve the model

The conversation will continue with the answer you choose.

Who was the first person to land on the moon?

Original Answer

The first person to set foot on the moon was Neil Armstrong, an American astronaut. He made history on July 20, 1969, during NASA's Apollo 11 mission, when he descended from the lunar module and took his famous steps onto the lunar surface. His iconic words as he stepped onto the moon were, "That's one small step for [a] man, one giant leap for mankind." Armstrong was followed by Buzz Aldrin, another astronaut from the same mission, making him the second person to walk on the moon.

Original answer is better

New Answer

The first person to set foot on the moon was astronaut Neil Armstrong. He was a member of the Apollo 11 mission, which was NASA's historic lunar landing mission. On July 20, 1969, Neil Armstrong and Buzz Aldrin became the first and second humans to walk on the moon while their colleague Michael Collins orbited above in the command module. Neil Armstrong's famous words as he stepped onto the lunar surface were, "That's one small step for man, one giant leap for mankind."

New answer is better

Neither answer is better Skip this step

Step 2

Collect comparison data, and train a reward model.

A prompt and several model outputs are sampled.



Figure: <https://medium.com/@joalages/direct-preference-optimization-dpo-622fc1f18707>

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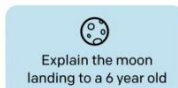
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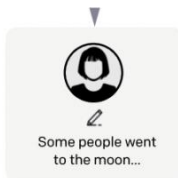
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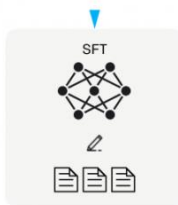
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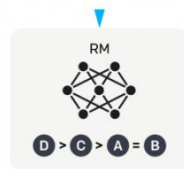
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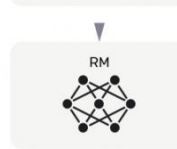
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Fine-tuning: Objectives

■ Instruction-tuning

- Improving the capability of the model in
 - specific tasks
 - following instructions specified in the prompt, thus making models more useful for practical use
 - e.g., taking advantage of the power of prompting and in-context learning helping adapt pre-trained models for practical use

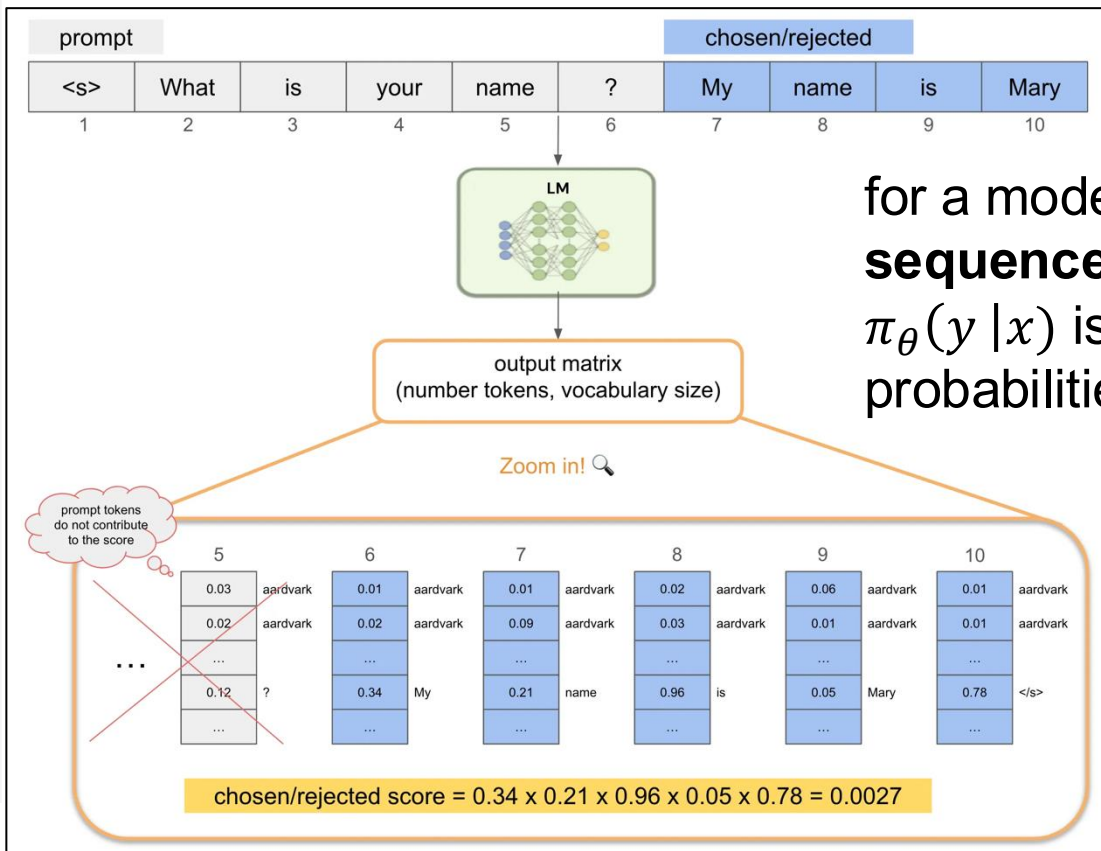
■ Preference tuning

- Aligning the model outputs to human preferences
 - E.g., safe and unbiased content

Fine-tuning: Techniques

- Supervised fine-tuning (SFT)
 - Based on `<prompt-output>` pairs, where output is a good output for the prompt according to “humans”
 - Requires hard-to-collect quality data
- Alignment with preference models
 - Based on `<prompt-output1-output2>` triples, where `output1` is better than `output2` according to labelers
 - Can use easier to collect data, e.g., users’ preferences over multiple generations
 - Different methods: based on reinforcement learning (RLHF) or RL-free (e.g., Direct Preference Optimization)

Input-conditioned Output Probabilities



for a model with parameters θ and a **sequence of tokens** y , $\pi_{\theta}(y | x)$ is the product of their probabilities, given the previous tokens x

RLHF: Reinforcement Learning with Human Feedback

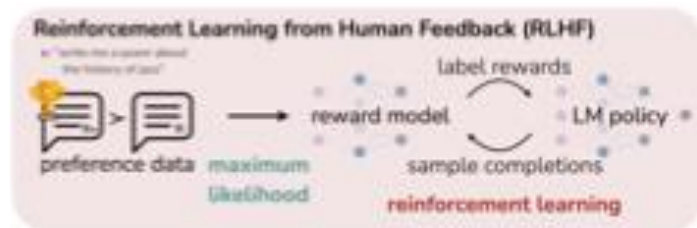


Image from DPO paper
(<https://arxiv.org/pdf/2305.18290>)

π_θ = the policy that we are training, $\sim$ the LLM we are optimizing; remark: $\pi_\theta(y|x) \in [0,1]$

π_{ref} = the policy before training, typically = π_{SFT}

RLHF optimization
objective for π_θ

$$\max_{\pi_\theta} \mathbb{E}_{x \sim \mathcal{D}, y \sim \pi_\theta(y|x)} [r_\phi(x, y)] - \beta \text{D}_{KL} [\pi_\theta(y | x) || \pi_{ref}(y | x)]$$

→ Requires reward model r_ϕ

Updates defined by RL algorithms, e.g., PPO Maximize over the policy π_θ : the expected reward $r_\phi(x, y)$, where prompts x are sampled from the dataset $\mathcal{D}$ and responses y are sampled from the policy $\pi_\theta(y|x)$, minus beta times the KL divergence between the current policy $\pi_\theta(y|x)$ and the reference policy $\pi_{ref}(y|x)$.

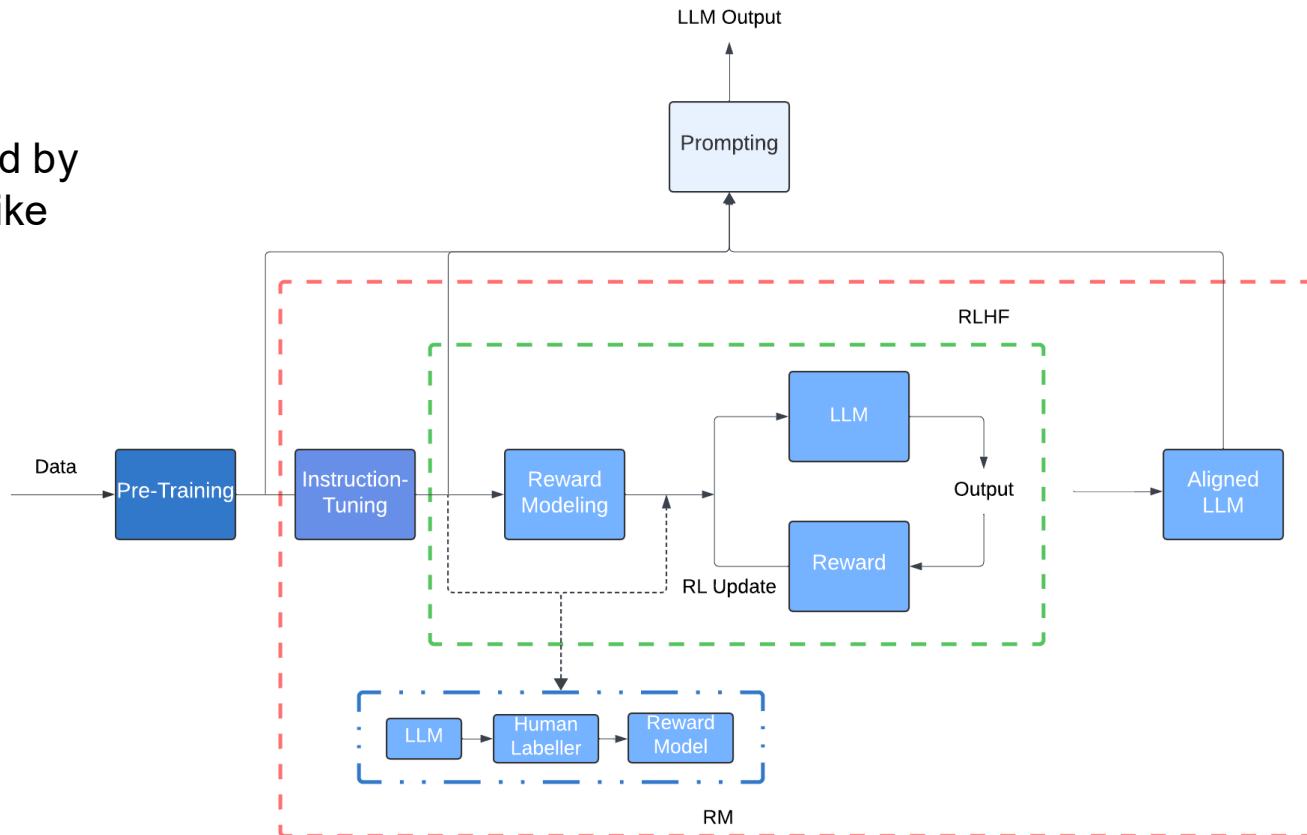
In practice: Maximize reward given by the reward model r_ϕ without diverging too much from the output of the reference model

We skip details about DPO ... it has the advantage that does not require a reward model and is more resource efficient

From Pre-train to Utilization

Remark:

RLHF can be replaced by
RL-free approaches like
DPO



A Practical Take on Model Optimization



Evals

Systematically measure performance.



Prompt engineering

Give context, instructions, and goals.



Fine-tuning

Train models to excel at a task.

- Check the OpenAI platform website
 - <https://platform.openai.com/docs/guides/model-optimization>
 - The website also provide a code-oriented guide to setting up Evals and Fine-tuning
- Prompt engineering is “easy”. You can check several guides online, for example the official guide by OpenAI
 - https://cookbook.openai.com/examples/gpt4-1_prompting_guide

LLM Training Wrap-up

■ Pre-training

- Language modeling on massive amount of data
- ... could be continued with more data: continual pre-training

Learn probability distributions from massive amount of data

■ Fine-tuning

- Supervised fine-tuning (SFT)
- Alignment with preference models
 - RLHF: preference data + reward model
 - DPO: preference data

Learn to:

- follow instructions
- align with user preferences and values

A Few More Things Useful for Data Enrichment

- **Multimodality**
 - Models can consider different modalities for input/output
- **Reasoning models (latest generation models)**
 - Models generating intermediate conclusions before returning an answer
- **Access to external knowledge**
 - We can integrate external knowledge by retrieving additional content: retrieval augmented generation
- **Tool usage**
 - The models can be prompted or fine-tuned to generate calls to external tools
- **Agentic frameworks**
 - We can setup multiple agents, where each agent can be specialized in one task
- **Not much time for details but it's good to know what these things are about**
 - Back to this after we see limitations of LLMs